

Forest Guardian AI

AI-Powered Forest Incident Reporting & Risk Assessment Platform

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Project Type: Engineering Internship / Final Project

Technology Stack: React, Vite, Tailwind CSS, n8n, OpenAI, Google Sheets, Power BI,
GitHub, Netlify

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Abstract

Forest Guardian AI is an AI-powered environmental monitoring platform designed to simplify and accelerate the reporting of forest-related incidents. Citizens can report incidents such as illegal logging, forest fires, wildlife poaching, pollution, and habitat destruction through a user-friendly web application. The submitted reports are automatically processed using an AI model that classifies the incident, assesses its severity, generates a risk score, and recommends appropriate actions. The analyzed data is stored in Google Sheets, visualized through a Power BI dashboard, and can trigger automated email notifications to the concerned authorities. The project demonstrates how AI and workflow automation can support environmental conservation and improve decision-making.

1. Introduction

Environmental degradation has become one of the major global concerns. Forests face continuous threats due to illegal logging, forest fires, wildlife poaching, pollution, and human encroachment. Traditional reporting methods are often slow, manual, and unstructured, making it difficult for authorities to respond quickly.

Forest Guardian AI addresses this problem by providing a digital platform that combines Artificial Intelligence, workflow automation, and data visualization to streamline incident reporting and improve response efficiency.

2. Problem Statement

Current forest incident reporting systems have several limitations:

- Manual reporting through phone calls or emails
- Slow response time
- Lack of centralized data management
- No automated risk assessment
- Difficulties in prioritizing incidents
- Limited real-time monitoring and analytics

These challenges reduce the effectiveness of environmental protection efforts.

3. Objectives

The objectives of the project are:

- Develop a web application for reporting forest incidents.
- Automate incident processing using AI.
- Classify reported incidents accurately.
- Generate risk scores and recommendations.
- Store reports in a centralized Google Sheet.
- Visualize insights using Power BI.
- Support faster decision-making for authorities.

4. Proposed Solution

Forest Guardian AI provides an end-to-end automated solution where users submit incident reports through a web application. The submitted information is sent to an n8n workflow using a webhook.

The AI agent analyzes the report and generates:

- Incident Type
- Risk Level
- Priority
- Risk Score
- Confidence Score
- Environmental Impact
- Recommendation
- Response Time
- Summary

The processed information is stored automatically in Google Sheets. Power BI reads this data to generate interactive dashboards, while email notifications can alert the relevant authorities.

5. System Architecture

Citizen
Forest Guardian AI Website
n8n Webhook
OpenAI AI Agent
Google Sheets Database
Power BI Dashboard
Forest Department

Data flows sequentially from citizen submission through automated AI analysis to the final dashboard used by the Forest Department for decision-making.

6. Technology Stack

Component	Technology
Frontend	React.js
Build Tool	Vite
Styling	Tailwind CSS
Automation	n8n
Artificial Intelligence	OpenAI GPT
Database	Google Sheets
Dashboard	Power BI
Version Control	GitHub
Deployment	Netlify

7. Project Workflow

- 1 Citizen visits the Forest Guardian AI website.
- 2 User fills the incident report form.
- 3 Data is sent to the n8n webhook.
- 4 The AI agent analyzes the incident.
- 5 AI generates risk assessment and recommendations.
- 6 Data is stored in Google Sheets.
- 7 Power BI updates the dashboard automatically.
- 8 Email alerts are sent to authorities when required.

8. Key Features

- User-friendly reporting interface
- AI-powered incident classification
- Automatic risk assessment
- Environmental impact analysis
- Priority-based recommendations
- Google Sheets integration
- Interactive Power BI dashboard
- Automated workflow using n8n
- Responsive web design
- Real-time data updates

9. AI Analysis

The OpenAI AI agent processes each incident report and generates structured outputs including:

- Incident Type
- Risk Level
- Priority

- Risk Score
- Confidence Level
- Environmental Impact
- Possible Causes
- Recommendation
- Response Time
- Summary

This enables authorities to prioritize critical incidents and take appropriate action.

10. Dashboard

Power BI provides interactive dashboards for monitoring reported incidents.

Dashboard Highlights:

- Total Reports
- Average Risk Score
- High-Risk Incidents
- Incident Type Distribution
- Location Analysis
- Time-based Trends
- Interactive Filters

The dashboard enables data-driven decision-making for forest authorities.

11. Results

The developed system successfully:

- Automates incident reporting
- Reduces manual processing
- Provides AI-generated risk assessment
- Centralizes environmental data
- Improves reporting efficiency
- Supports faster decision-making

- Enhances data visualization using Power BI

12. Future Scope

Future improvements include:

- Mobile Application
- GPS Integration
- Satellite Image Monitoring
- AI Image Recognition
- Drone-based Forest Surveillance
- Real-time GIS Mapping
- SMS Notifications
- Multi-language Support
- Government Portal Integration
- Predictive Risk Analysis

13. SDG Alignment

This project contributes to United Nations Sustainable Development Goal 15 (Life on Land) by promoting responsible forest management, protecting biodiversity, improving environmental monitoring, and supporting sustainable ecosystem conservation through artificial intelligence and automation.

14. Conclusion

Forest Guardian AI demonstrates how Artificial Intelligence, workflow automation, and data visualization can modernize environmental incident reporting. By integrating React, n8n, OpenAI, Google Sheets, and Power BI, the system provides an efficient platform for reporting, analyzing, and monitoring forest-related incidents. The project reduces manual effort, improves response prioritization, and supports informed decision-making, making it a scalable solution for environmental conservation.

GitHub Repository

<https://github.com/patilyash8113/Forest-guardian-AI>

Live Website

<https://forest-guardian-ai.netlify.app/>

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